

Requirements Engineering - Problem Statement #58

Digital Art Commission & Watermarking Portal

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Apply a dynamic diagonal overlay watermark to every artist WIP before client display.	High	Pass: watermarked draft shown; Fail: unwatermarked draft exposed.	Protects WIP artwork.
FR-002	Functional	Allow the Client Buyer to view watermarked WIP and provide feedback or approval.	High	Feedback/approval is stored against the commission.	Supports iterative review.
FR-003	Functional	Allow the Client Buyer to complete the required milestone payment.	High	Successful payment is recorded; failed/pending payment is not treated as paid.	Payment gates final release.
FR-004	Functional	Unlock the final high-resolution source file only after successful payment verification.	High	Final asset is unavailable before verified payment and available after it.	Prevents premature release.
FR-005	Functional	Allow an authorized client to download the final asset using a secure, time-limited link.	High	Valid authorized link works; unauthorized/expired link fails.	Protects final delivery.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Final asset downloads use signed S3 URLs valid for 60 minutes only.	High	Tests confirm link generation and expiry after 60 minutes.	Limits exposure window.
NFR-002	Security	Commission data/assets are restricted to authenticated and authorized users associated with the commission.	High	Unauthorized and unauthenticated requests cannot access private assets.	Protects artwork and commission data.

Requirements Traceability Matrix (RTM)

Req.	Summary	Related Use Case / Flow	Primary Component(s)	Verification / Evidence
FR-001	Watermark WIP	Upload WIP -> Apply Dynamic Watermark	Watermark Service; Asset Storage	Verify watermark is visible in client portal.
FR-002	Review WIP	Review WIP; Feedback / Approval	Portal; Order Manager	Submit feedback/approval and verify storage.
FR-003	Milestone payment	Make Payment -> Verify Payment	Payment Service; Order Manager	Successful payment changes milestone to paid.
FR-004	Payment-gated unlock	Verify Payment -> Unlock Final Asset	Order Manager; Secure Delivery	Compare access before and after verified payment.
FR-005	Secure download	Download Final Asset -> Signed URL	Secure Delivery; Portal	Authorized user receives valid time-limited link.
NFR-001	60-minute URL	Generate Signed URL	Secure Delivery	Validate within window; reject after expiry.
NFR-002	Authorized access	Authentication / authorization flow	Portal; Order Manager; Secure Delivery	Test authorized, unauthorized and unauthenticated requests.